

Class 10: Structural Bioinformatics 1

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Base** (PDB).

PDB Statistics

What is in teh PDB in terms of molecule type and structure determination method?

```
pdb<-read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1

	Neutron	Other	Total
1	84	32	217,375
2	1	0	14,283
3	0	0	16,277
4	3	1	5,004
5	0	0	244
6	0	4	22

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy?

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

```
pdb$EM
```

```
[1] "23,111" "3,741" "6,751" "250" "27" "0"
```

This print out above `pdb$x.ray` is “character” and not “numeric”. Therefore I cannot do math with it.

Two functions that can help are `sub()` and `as.numeric()`

```
# We want to get rid (or sub out) commas:
x<- pdb$X.ray
tmp<-sub(",", "", x = pdb$X.ray)
sum(as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do this:

```
rm.comma<-function(x){
  tmp<-sub(",", "", x = pdb$EM)
  sum(as.numeric(tmp))
}
```

```
n.tot<-rm.comma(pdb$Total)
n.xray<-rm.comma(pdb$X.ray)
n.em <-rm.comma(pdb$EM)
```

We could also use a different import function for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file)

```
pdb$Neutron
```

```
[1] 84  1  0  3  0  0
```

```
library(readr)
read.csv("Data Export Summary.csv")
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					
1	84 32	217,375				
2	1 0	14,283				
3	0 0	16,277				
4	3 1	5,004				
5	0 0	244				
6	0 4	22				

```
pdb$Total <- as.numeric(gsub(",", "", pdb$Total))
pdb$X.ray <- as.numeric(gsub(",", "", pdb$X.ray))
pdb$EM <- as.numeric(gsub(",", "", pdb$EM))
```

```
n.tot <- sum(pdb$Total)
n.xray <- sum(pdb$X.ray)
n.em <- sum(pdb$EM)
```

```
(n.xray / n.tot) * 100
```

```
[1] 80.48577
```

```
(n.em / n.tot) * 100
```

```
[1] 13.38046
```

Q2: What proportion of structures in the PDB are protein?

```
pdb$Total[1]
```

```
[1] 217375
```

The total number of protein sequences in UniProt is 202,556,314

```
(217375/202556314) *100
```

```
[1] 0.1073158
```

Key-point: we have very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-Ray crystallography)

Q3: Professor said to skip

Visualizing PDB Data with Mol-Star

Main stand alone web version with all features is at <https://molstar.org/viewer/>

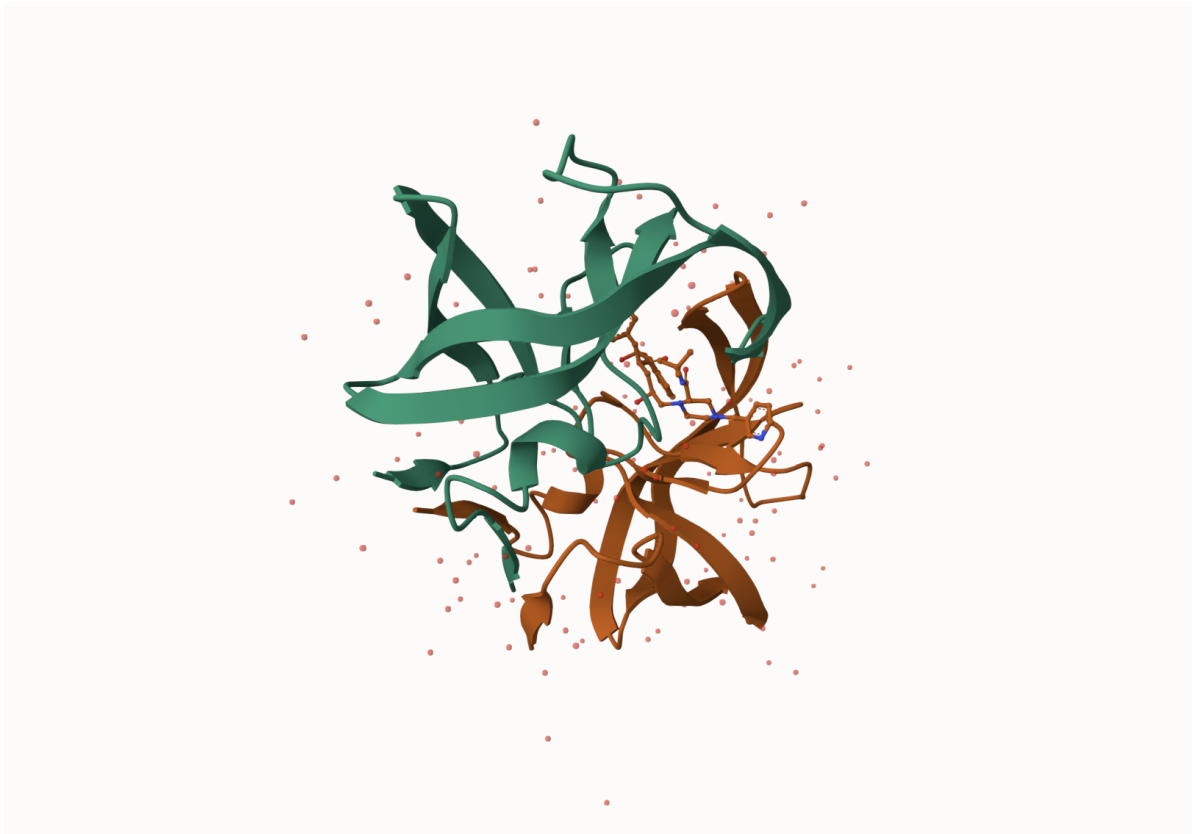
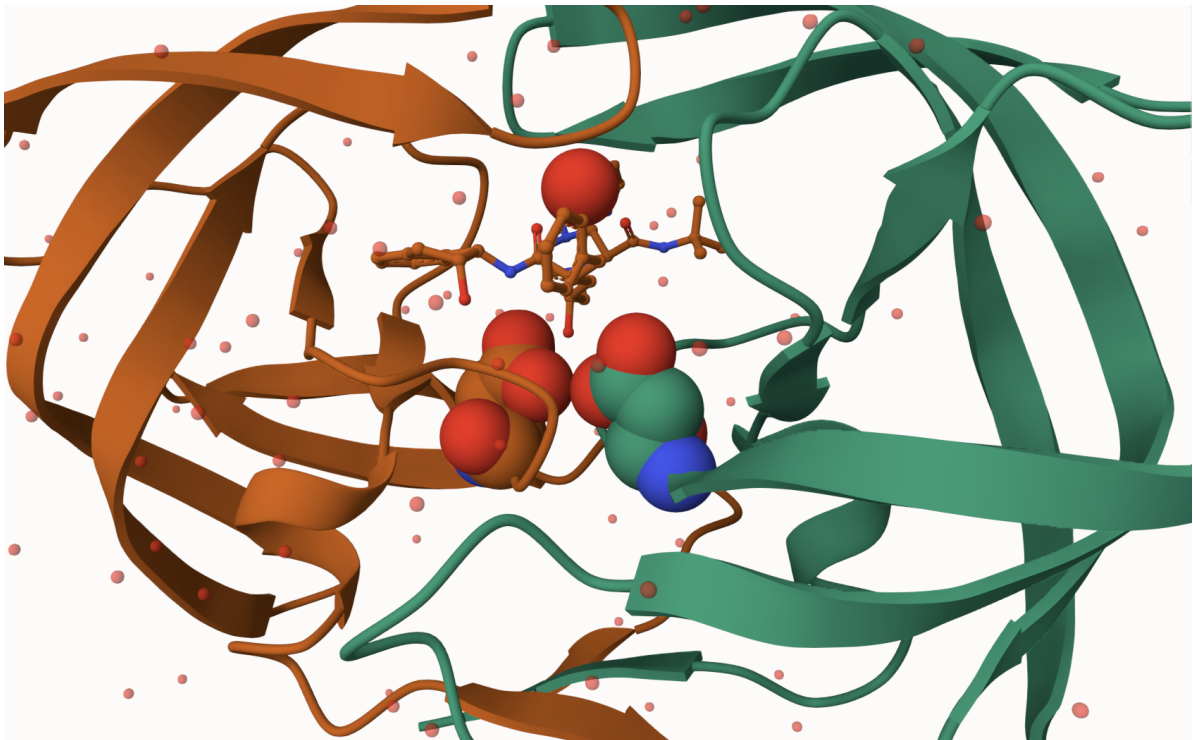
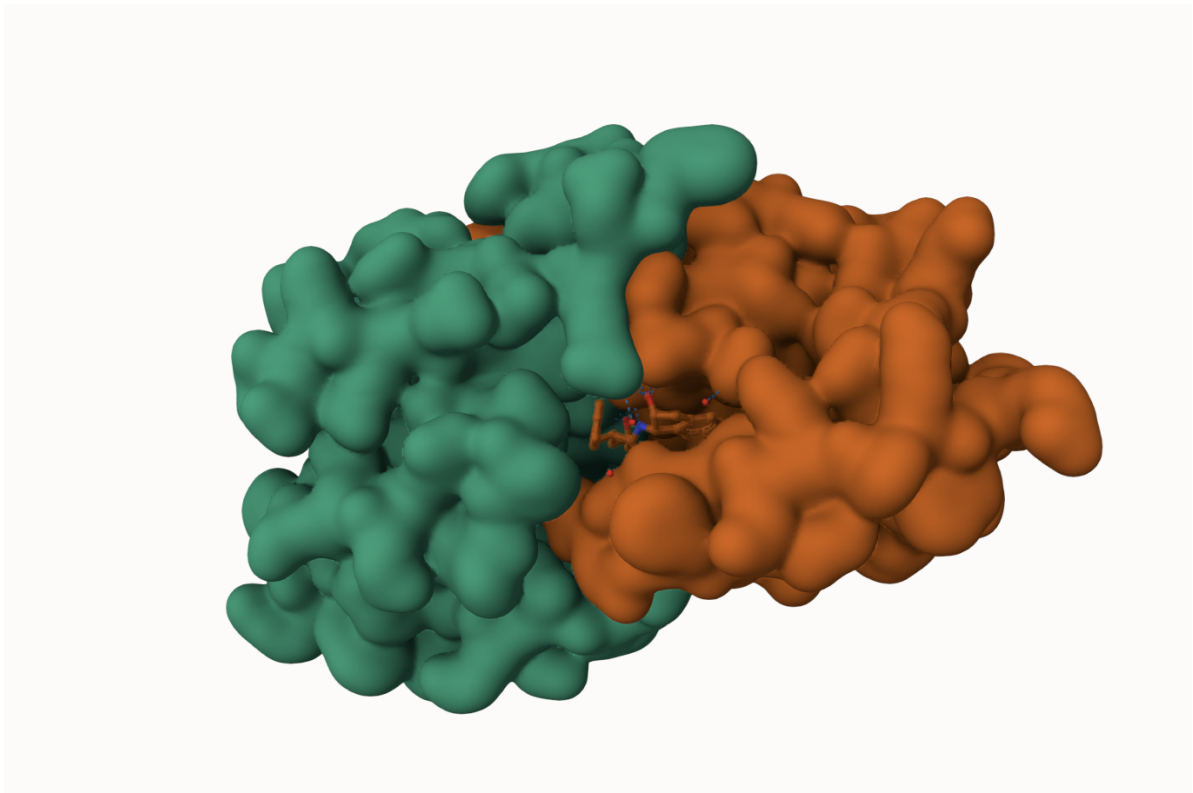


Figure 1: The HIV-protease enzyme is a homodimer with two chains



Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

The only molecule detected in the image is the oxygen molecule of water. We only see one atom because hydrogen atoms are extremely small and have a very low electron density, as they only have one electron.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

The residue number of this water molecule is 308.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

The figure is above.

Getting started with the Bio3D package

Bio3D is an R package from CRAN for structural bioinformatics

```
library(bio3d)
pdb<-read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
  Total Atoms#: 1686,  XYZs#: 5058  Chains#: 2  (values: A B)

Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172  (residues: 128)
```

Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues.

Q8: Name one of the two non-protein residues?

One of the two non-protein residues is HOH (water)

Q9: How many protein chains are in this structure?

There are two protein chains in this structure.

```
attributes(pdb)
```

```
$names
```

```
[1] "atom"    "xyz"      "seqres"   "helix"    "sheet"    "calpha"   "remark"   "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

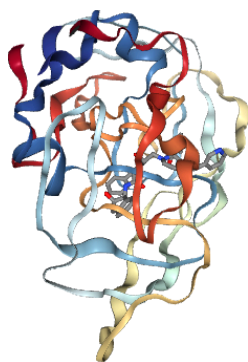
	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40
	segid	elesy	charge										
1	<NA>	N	<NA>										
2	<NA>	C	<NA>										


```
3 <NA>      C  <NA>
4 <NA>      O  <NA>
5 <NA>      C  <NA>
6 <NA>      C  <NA>
```

```
head(pdbseq(pdb))
```

```
  1    2    3    4    5    6
"P" "Q" "I" "T" "L" "W"
```

```
library(bio3dview)
view.pdb(pdb)
```



Q10. Which of the packages above is found only on BioConductor and not CRAN?

The msa package is found only on BioConductor and not CRAN.

Q11. Which of the above packages is not found on BioConductor or CRAN?:

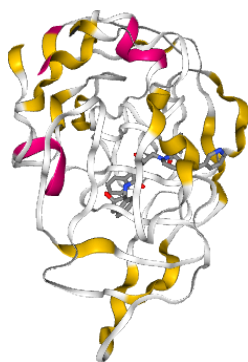
The bio3dview package is not found on BioConductor or CRAN.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True.

Let's try a custom view

```
view.pdb(pdb,colorScheme="sse", backgroundColor = "Black")
```

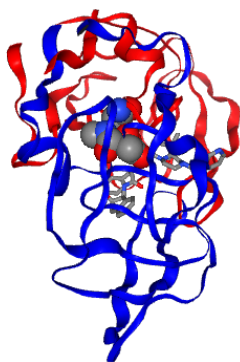


Q. create a custom view highlighting the active site ASP (`resno=25`), the two chains (in your choice of colors) and the ligand all on a custom color background.

```
library(NGLVieweR)

active.site<-atom.select(pdb,resno=25)

view.pdb(pdb, cols = c("red","blue"),highlight = active.site, highlight.style = "spacefill",
```



Predict the flexibility of a given structure

Let's do a normal mode analysis (NMA) to predict the flexibility of a give pdb object:

```
adk<-read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 244 (residues: 244)
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

Protein sequence:

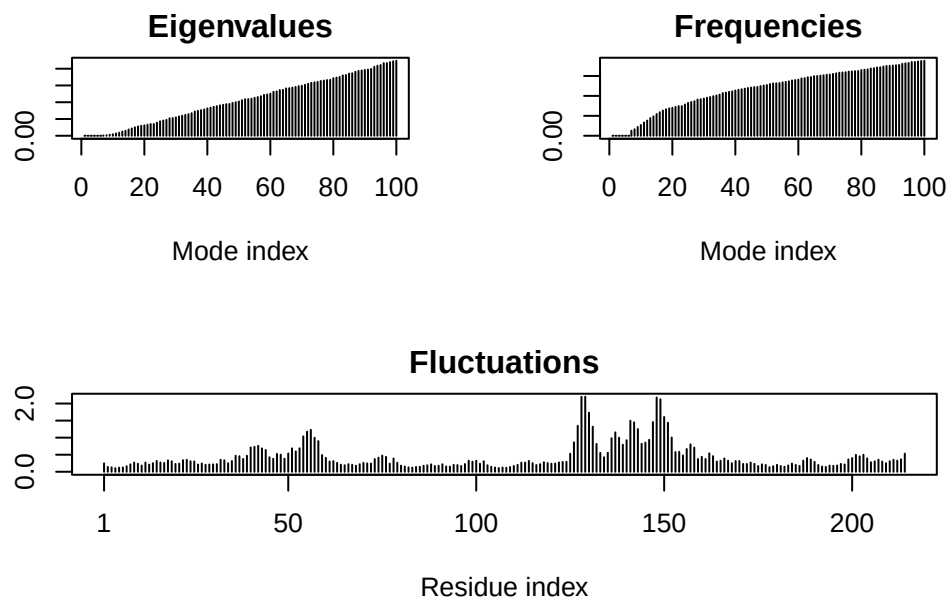
```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV
DELVIALVKERIAQEDCRNGFLLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM
TAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
m<-nma(adk)
```

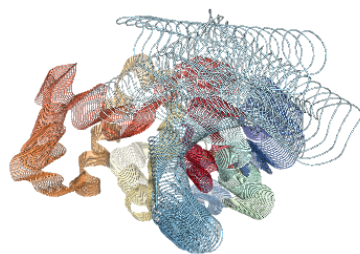
```
Building Hessian...      Done in 0.008 seconds.
Diagonalizing Hessian... Done in 0.181 seconds.
```

```
plot(m)
```



View the results with an interactive structue view:

```
view.nma(m)
```

Write out the results for viewing in Mol-star:

```
mktrj(m,file="nma.pdb")
```

Comparative structure analysis of Adenylate Kinase

Our first step is find a sequence for this family. We will use the database ID “1ake_A” here:

```
id<-"1ake_A"  
aa<-get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

```
aa
```

```
      1      .      .      .      .      .      .      60  
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLAAVKSGSELGKQAKDIMDAGKLV  
      1      .      .      .      .      .      .      60  
  
      61      .      .      .      .      .      .      120  
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVDRI  
      61      .      .      .      .      .      .      120  
  
      121     .      .      .      .      .      .      180  
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
      121     .      .      .      .      .      .      180  
  
      181     .      .      .      214  
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG  
      181     .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

1 sequence rows; 214 position columns (214 non-gap, 0 gap)

+ attr: id, ali, call

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

There are 214 amino acids in this sequence, which also makes up the length of this sequence.

Search for related sequences in the database

```
blast<-blast.pdb(aa)
```

Searching ... please wait (updates every 5 seconds) RID = ZZNG1ZME014

.....

Reporting 96 hits

```
head(blast$hit.tbl)
```

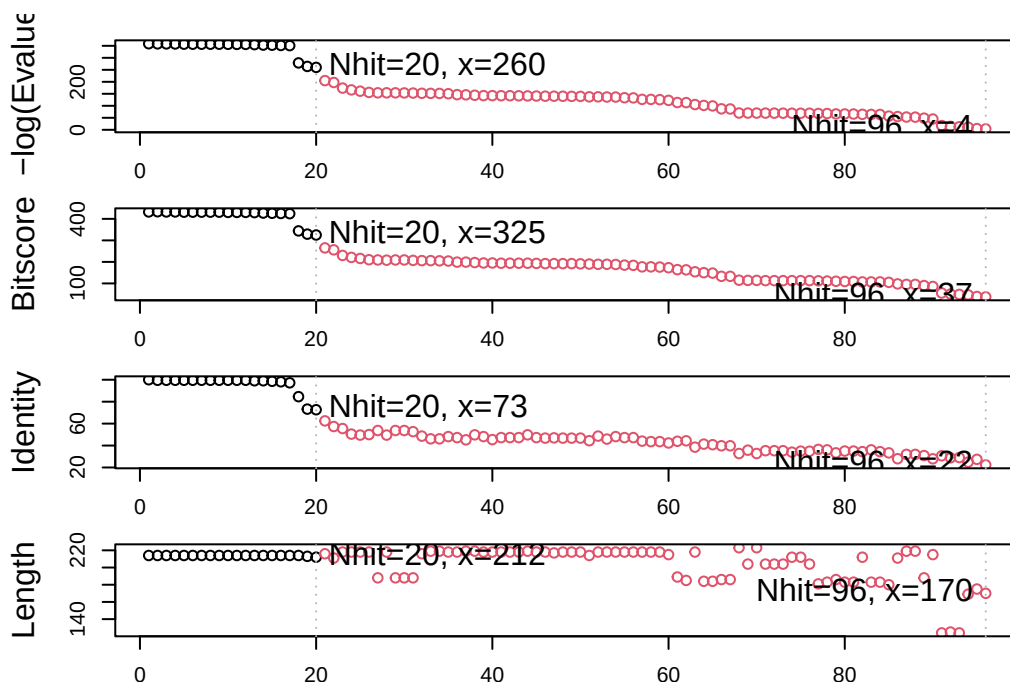
	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_6117585	1AKE_A	100.000	214	0	0	1
2	Query_6117585	8BQF_A	99.533	214	1	0	1
3	Query_6117585	4X8M_A	99.533	214	1	0	1
4	Query_6117585	6S36_A	99.533	214	1	0	1
5	Query_6117585	9R6U_A	99.533	214	1	0	1
6	Query_6117585	9R71_A	99.533	214	1	0	1

	q.end	s.start	s.end	evalue	bitscore	positives	mlog.evalue	pdb.id	acc
1	214	1	214	1.82e-156	432	100.00	358.6044	1AKE_A	1AKE_A
2	214	21	234	2.97e-156	433	100.00	358.1147	8BQF_A	8BQF_A
3	214	1	214	3.25e-156	432	100.00	358.0246	4X8M_A	4X8M_A
4	214	1	214	4.77e-156	432	100.00	357.6409	6S36_A	6S36_A
5	214	1	214	1.06e-155	431	99.53	356.8424	9R6U_A	9R6U_A
6	214	1	214	1.25e-155	431	99.53	356.6775	9R71_A	9R71_A

```
hits<-plot(blast)
```

* Possible cutoff values: 260 3
Yielding Nhits: 20 96

* Chosen cutoff value of: 260
Yielding Nhits: 20



```
hits$pdb.id
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
[9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
files <- get.pdb(hits$pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6S36.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

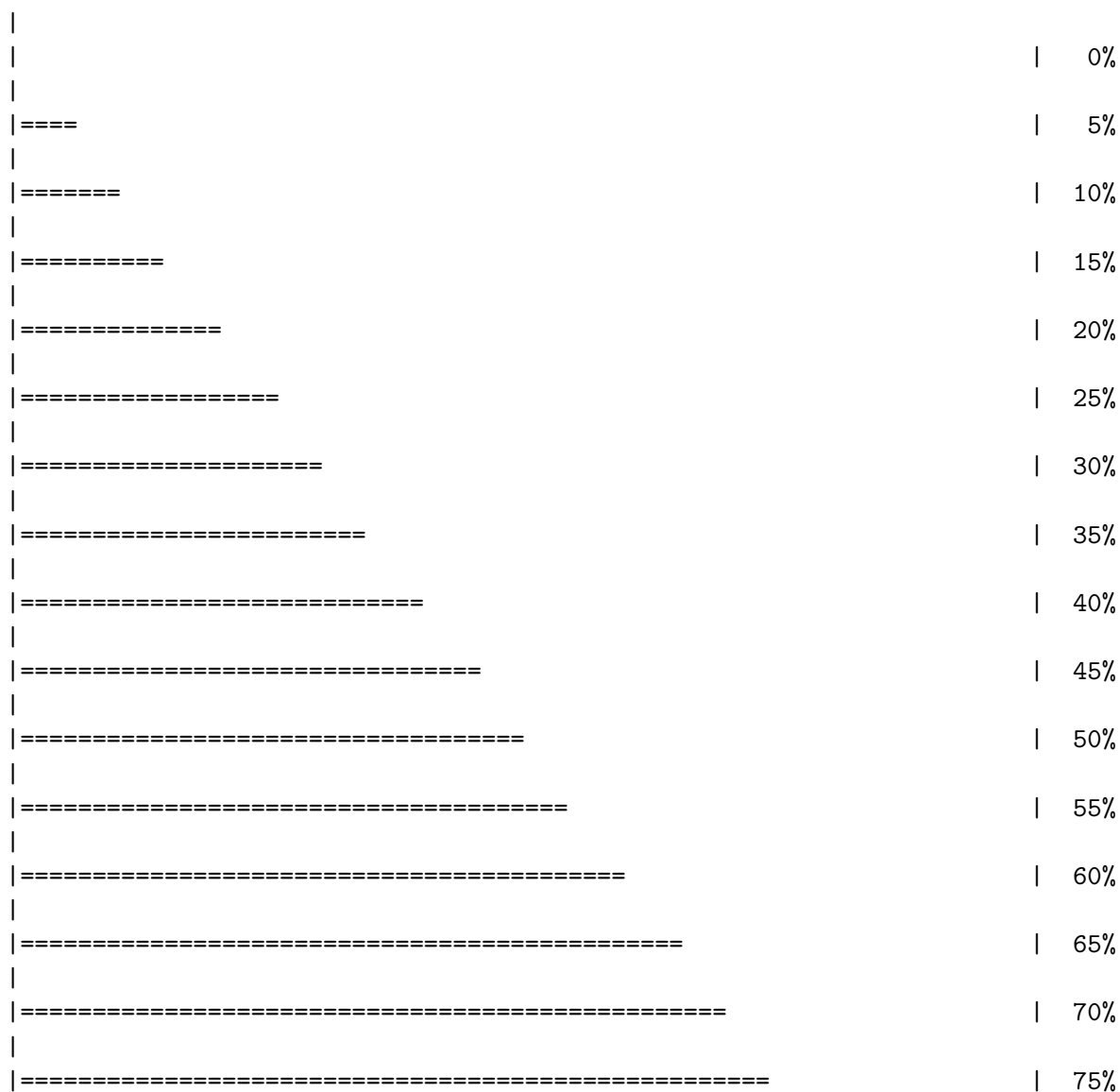
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download



```

|
|=====| 80%
|
|=====| 85%
|
|=====| 90%
|
|=====| 95%
|
|=====| 100%

```

Align and superimpose all these ADK structures

```
pdbbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```

pdbbs/split_chain/1AKE_A.pdb
pdbbs/split_chain/8BQF_A.pdb
pdbbs/split_chain/4X8M_A.pdb
pdbbs/split_chain/6S36_A.pdb
pdbbs/split_chain/9R6U_A.pdb
pdbbs/split_chain/9R71_A.pdb
pdbbs/split_chain/8Q2B_A.pdb
pdbbs/split_chain/8RJ9_A.pdb
pdbbs/split_chain/6RZE_A.pdb
pdbbs/split_chain/4X8H_A.pdb
pdbbs/split_chain/3HPR_A.pdb
pdbbs/split_chain/1E4V_A.pdb
pdbbs/split_chain/5EJE_A.pdb
pdbbs/split_chain/1E4Y_A.pdb
pdbbs/split_chain/3X2S_A.pdb
pdbbs/split_chain/6HAP_A.pdb
pdbbs/split_chain/6HAM_A.pdb
pdbbs/split_chain/8PVW_A.pdb
pdbbs/split_chain/4K46_A.pdb
pdbbs/split_chain/4NP6_A.pdb

```

```

PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
.. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE
. PDB has ALT records, taking A only, rm.alt=TRUE

```

```

.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

```

Extracting sequences

```

pdb/seq: 1   name: pdbc/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbc/split_chain/8BQF_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbc/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbc/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbc/split_chain/9R6U_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbc/split_chain/9R71_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbc/split_chain/8Q2B_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbc/split_chain/8RJ9_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbc/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pdbc/split_chain/4X8H_A.pdb
pdb/seq: 11  name: pdbc/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbc/split_chain/1E4V_A.pdb
pdb/seq: 13  name: pdbc/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14  name: pdbc/split_chain/1E4Y_A.pdb
pdb/seq: 15  name: pdbc/split_chain/3X2S_A.pdb
pdb/seq: 16  name: pdbc/split_chain/6HAP_A.pdb
pdb/seq: 17  name: pdbc/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18  name: pdbc/split_chain/8PVW_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19  name: pdbc/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE

```


pdb/seq: 20 name: pdb/split_chain/4NP6_A.pdb

pdb/s

	1	.	.	.	40
[Truncated_Name:1] 1AKE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:2] 8BQF_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:3] 4X8M_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:4] 6S36_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:5] 9R6U_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:6] 9R71_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:7] 8Q2B_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:8] 8RJ9_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:9] 6RZE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:10] 4X8H_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:11] 3HPR_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:12] 1E4V_A.pdb	--MRIILLGAPVAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:13] 5EJE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:14] 1E4Y_A.pdb	--MRIILLGALVAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:15] 3X2S_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:16] 6HAP_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:17] 6HAM_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:18] 8PVW_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:19] 4K46_A.pdb	--MRIILLGAPGAGKGTQAQFIMAKFGIPQISTG	DMLRAA			
[Truncated_Name:20] 4NP6_A.pdb	NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTG	DMLRAA			
	*****	*****	*~*****		
	1	.	.	.	40
	41	.	.	.	80
[Truncated_Name:1] 1AKE_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:2] 8BQF_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQE---				
[Truncated_Name:3] 4X8M_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:4] 6S36_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:5] 9R6U_A.pdb	VKSGSELGAQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:6] 9R71_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:7] 8Q2B_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:8] 8RJ9_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:9] 6RZE_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:10] 4X8H_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:11] 3HPR_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:12] 1E4V_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR				
[Truncated_Name:13] 5EJE_A.pdb	VKSGSELGKQAKDIMDACKLVDELVIALVKERIAQEDCR				

[Truncated_Name:14] 1E4Y_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR	
[Truncated_Name:15] 3X2S_A.pdb	VKSGSELGKQAKDIMDCGKLVDELVIALVKERIAQEDSR	
[Truncated_Name:16] 6HAP_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVRERICQEDSR	
[Truncated_Name:17] 6HAM_A.pdb	IKSGSELGKQAKDIMDAGKLVDEIIIALVKERICQEDSR	
[Truncated_Name:18] 8PVW_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR	
[Truncated_Name:19] 4K46_A.pdb	IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA	
[Truncated_Name:20] 4NP6_A.pdb	IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE	
	^* *^*** *** ^~* **~*~~~~*~^*** *	
	41 . . .	80
	81 . . .	120
[Truncated_Name:1] 1AKE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:2] 8BQF_A.pdb	-GFLLDGFPR TIPQADAMKEAGINVDYVIEFDVPDELIVD	
[Truncated_Name:3] 4X8M_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:4] 6S36_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:5] 9R6U_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:6] 9R71_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDALIVD	
[Truncated_Name:7] 8Q2B_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:8] 8RJ9_A.pdb	NGFLLAGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:9] 6RZE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:10] 4X8H_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:11] 3HPR_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:12] 1E4V_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:13] 5EJE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:14] 1E4Y_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:15] 3X2S_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:16] 6HAP_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:17] 6HAM_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:18] 8PVW_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	
[Truncated_Name:19] 4K46_A.pdb	KGFLLDGFPR TIPQADGLKEVGVVVDYVIEFDVADSVIVE	
[Truncated_Name:20] 4NP6_A.pdb	KGFLLDGFPR TIPQADGLKEMGINVDYVIEFDVADDVIVE	
	**** *****^~** *^ *****^***** * ^**^	
	81 . . .	120
	121 . . .	160
[Truncated_Name:1] 1AKE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:2] 8BQF_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:3] 4X8M_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:4] 6S36_A.pdb	KIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:5] 9R6U_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:6] 9R71_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	
[Truncated_Name:7] 8Q2B_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKA	
[Truncated_Name:8] 8RJ9_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	

[Truncated_Name:9] 6RZE_A.pdb	AIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:10] 4X8H_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:11] 3HPR_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDGTGEELTTRKD
[Truncated_Name:12] 1E4V_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:13] 5EJE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:14] 1E4Y_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:15] 3X2S_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:16] 6HAP_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:17] 6HAM_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:18] 8PVW_A.pdb	RILKR--GETSGRV-----D
[Truncated_Name:19] 4K46_A.pdb	RMAGRRAHLASGRTYHNVNPPKVEGKDDVTGEDLVIRED
[Truncated_Name:20] 4NP6_A.pdb	RMAGRRAHLPSGRTYHVNVNPPKVEGKDDVTGEDLVIRED
	^ * ***
	121 . . . 160
	161 . . . 200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:2] 8BQF_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:3] 4X8M_A.pdb	DQEETVRKRLVEWHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:4] 6S36_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:5] 9R6U_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:6] 9R71_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:7] 8Q2B_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:8] 8RJ9_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:9] 6RZE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:10] 4X8H_A.pdb	DQEETVRKRLVEYHQMTAALIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:11] 3HPR_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:12] 1E4V_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:13] 5EJE_A.pdb	DQEECVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:14] 1E4Y_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:15] 3X2S_A.pdb	DQEETVRKRLCEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:16] 6HAP_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:17] 6HAM_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:18] 8PVW_A.pdb	DNEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:19] 4K46_A.pdb	DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:20] 4NP6_A.pdb	DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
	* * * * * ^ * * * * * * * * * *
	161 . . . 200
	201 . 216
[Truncated_Name:1] 1AKE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:2] 8BQF_A.pdb	TKPVAEVRADLEKIL-
[Truncated_Name:3] 4X8M_A.pdb	TKPVAEVRADLEKILG

```

[Truncated_Name:4] 6S36_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:5] 9R6U_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:6] 9R71_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:7] 8Q2B_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:8] 8RJ9_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:9] 6RZE_A.pdb      TKPVAEVRADLEKILG
[Truncated_Name:10] 4X8H_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:11] 3HPR_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:12] 1E4V_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:13] 5EJE_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:14] 1E4Y_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:15] 3X2S_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:16] 6HAP_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:17] 6HAM_A.pdb     TKPVCEVRADLEKILG
[Truncated_Name:18] 8PVW_A.pdb     TKPVAEVRADLEKILG
[Truncated_Name:19] 4K46_A.pdb     TKAVAEVSAELEKALA
[Truncated_Name:20] 4NP6_A.pdb     TKQVSEVSADIAKALA
                                ** * ** *~* *
                                201      .      216

```

Call:

```
pdaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

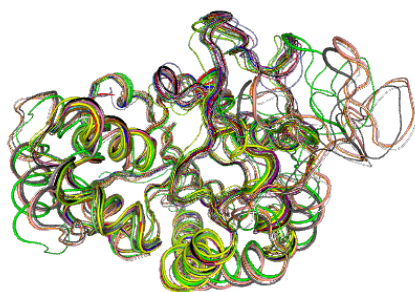
```
pdbs, fasta
```

Alignment dimensions:

```
20 sequence rows; 216 position columns (182 non-gap, 34 gap)
```

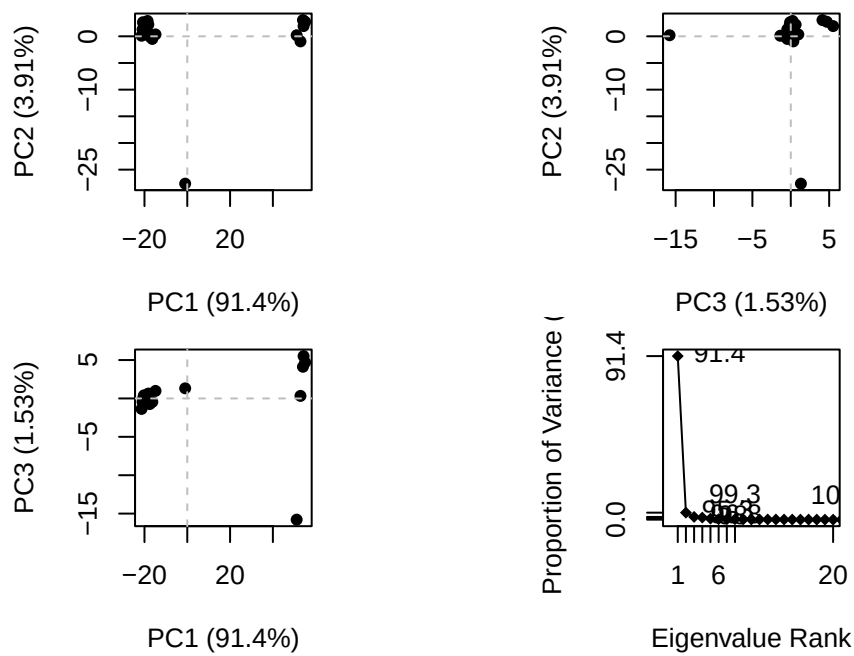
```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```
view.pdbs(pdbs)
```

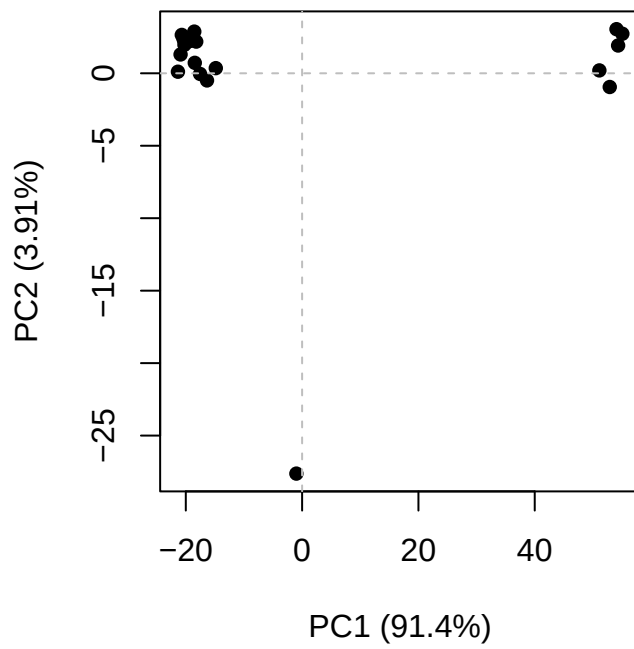


PCA of all this structural data (x,y, and z atom coordinates):

```
pc<-pca(pdbbs)
plot(pc)
```

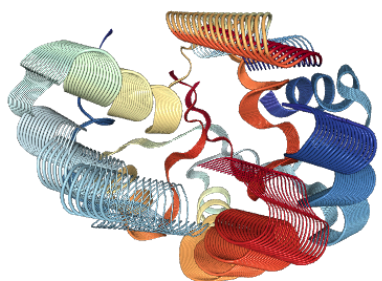


```
plot(pc,1:2)
```



Interactive View of the MC1 captured structural differences:

```
view.pca(pc)
```




```
mktrj(pc,file="pca.pdb")
```

```
ids <- basename.pdb(pdb$ids)
```

```
anno <- pdb.annotate(ids)
```

```
unique(anno$source)
```

```
[1] "Escherichia coli"  
[2] "Escherichia coli K-12"  
[3] "Escherichia coli O139:H28 str. E24377A"  
[4] "Escherichia coli str. K-12 substr. MDS42"  
[5] "Photobacterium profundum"  
[6] "Vibrio cholerae O1 biovar El Tor str. N16961"
```

```
modes <- nma(pdb$ids)
```

Warning in nma.pdb(pdb\$ids): 8BQF_A.pdb might have missing residue(s) in structure:
Fluctuations at neighboring positions may be affected.

Details of Scheduled Calculation:

```
... 20 input structures  
... storing 540 eigenvectors for each structure  
... dimension of x$U.subspace: ( 546x540x20 )  
... coordinate superposition prior to NM calculation  
... aligned eigenvectors (gap containing positions removed)  
... estimated memory usage of final 'eNMA' object: 45.1 Mb
```

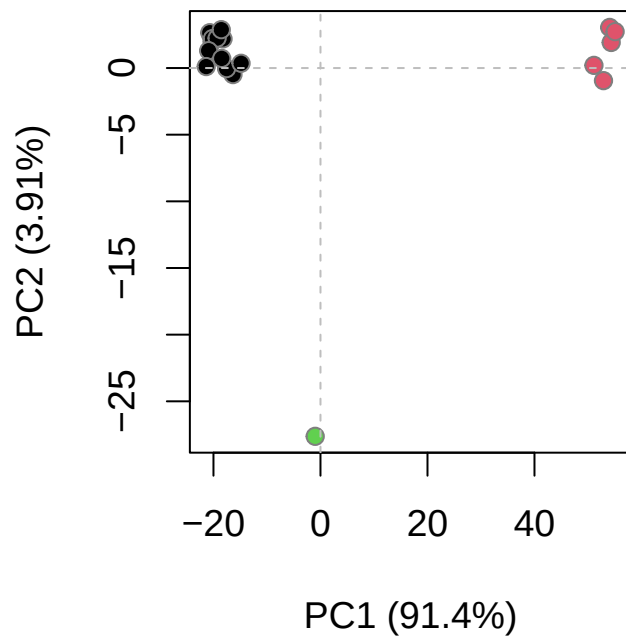
	0%
====	5%
=====	10%
=====	15%
=====	20%



```
rd <- rmsd(pdb)
```

Warning in rmsd(pdb): No indices provided, using the 182 non NA positions

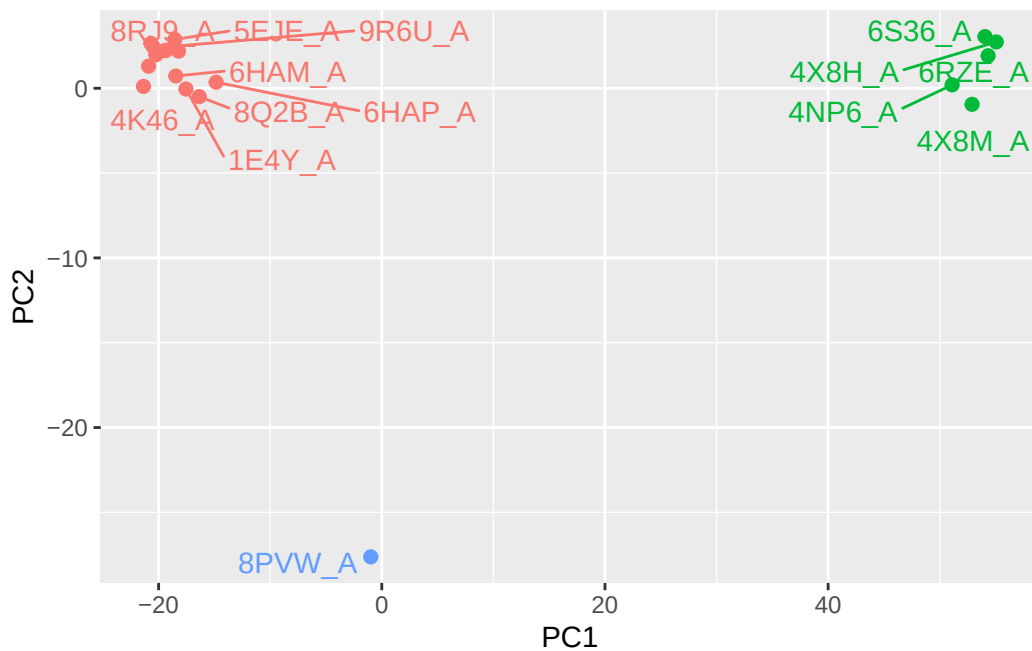
```
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)
pc.xray <- pca(pdb)
plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



```
library(ggplot2)
library(ggrepel)

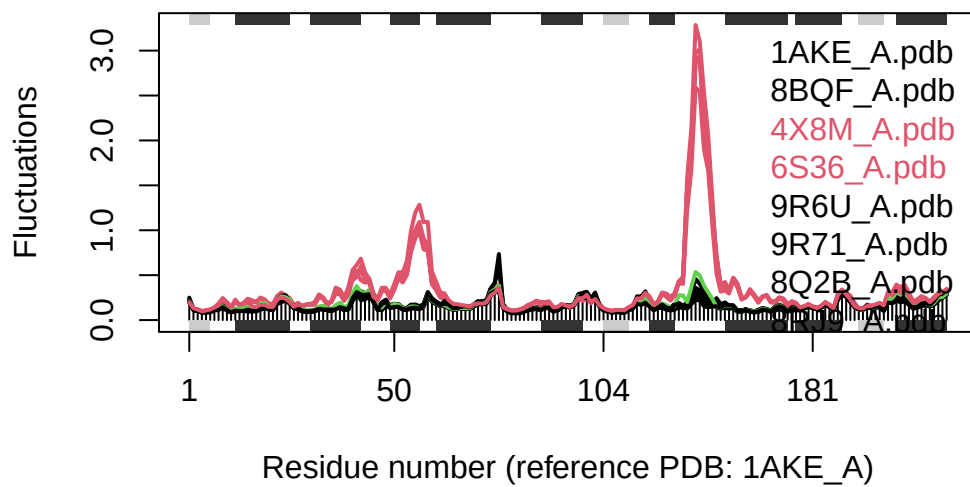
df <- data.frame(PC1=pc.xray$z[,1],
                 PC2=pc.xray$z[,2],
                 col=as.factor(grps.rd),
                 ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p
```



```
plot(modes, pdb, col=grps.rd)
```

Extracting SSE from pdb\$sse attribute



Q14. What do you note about this plot? Are the black and colored lines similar or different? Where do you think they differ most and why?

The black and colored lines are relatively similar because they tend to follow the same trend, but they differ most in their magnitude as the colored lines have a much larger peak than the black line. They differ the most around 140 and this could be due to conformational change of the protein.